

AI DATA CENTRE CAPACITY IN CANADA

HYPERSCALERS VERSUS AI-NATIVE PROVIDERS

*A Strategic, Power and Market Reference for Investors,
Operators and Boards*

B² ADVISORY

Independent Telecom & Infrastructure Advisory



FREE PREVIEW EDITION

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This preview contains the front matter, Executive Summary and Section 1 of the full 48-page report.

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- ✓ Included in this preview — Executive Summary
- ✓ Included in this preview — 1. Market Overview and Size
- 2. Segmenting the Market: Participants and the Value Chain
- 3. Who's Who: Participants by Segment
- 4. Scaling Compute Capacity
- 5. Contracting Capacity: Leases, Credit and Who Funds What
- 6. Power Procurement and Grid Access
- 7. Cooling, Density and Facility Design
- 8. Location Selection: Province by Province
- 9. Sovereignty as a Selection Criterion
- 10. Where the Two Models Converge
- 11. Key Considerations for Boards and Investment Committees
- 12. Risk Factors
- 13. Outlook and Catalysts to Watch
- 14. Key Questions Answered
- 15. Glossary of Terms
- Appendix A. Methodology and Sources
- Appendix B. About B² Advisory

Get the complete report

The full edition segments the market into five participant types across a five-layer value chain; compares the TELUS owner-operator and Bell landlord-and-channel models; profiles the hyperscalers, neoclouds, Canadian GPU clouds and AI labs active in Canada; and covers lease and credit structures, who funds what, the provincial power-access regimes of Alberta, Quebec, Ontario, British Columbia and Saskatchewan, liquid cooling and rack density, federal sovereignty programs, risk factors and the catalysts to watch — assembled for investors, executives and boards evaluating exposure to Canadian AI infrastructure.

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Executive Summary

Canada has moved quickly from a secondary cloud market to a contested destination for AI compute. Requested data-centre load in Alberta alone — about **19.6 GW** on the provincial grid operator's connection list — now exceeds the province's record peak demand of 12.8 GW. Ontario estimates that its own requests could exceed **10 GW**. Meta has broken ground on a gigawatt-scale campus in Alberta, Microsoft has committed roughly US\$5.4 billion (C\$7.5bn) to new Canadian capacity over two years, and Bell Canada and TELUS have each launched sovereign AI infrastructure programs.

Two broad kinds of buyer are competing for the same scarce megawatts. **Hyperscalers** — the hyperscale clouds (AWS, Microsoft, Google and Oracle) and self-use platforms such as Meta — plan multi-decade campuses and finance them from investment-grade balance sheets. **AI-native providers** — global neoclouds such as CoreWeave and Cerebras, Canadian GPU clouds such as TELUS's Sovereign AI Factory, Denvr Dataworks and BUZZ HPC, and the AI labs that buy their capacity — plan around GPU generations, chase power that can be energized quickly, and depend on landlords, lenders and anchor customers to fund growth. Within each group, roles differ markedly, so this Report segments participants into five types and places them, together with landlords, channels and power suppliers, on a five-layer value chain. It then compares how the groups scale compute, contract for power and cooling, and select locations, and sets out what that means for investors, operators and boards evaluating exposure to the Canadian market.

Key findings

- **Power, not capital or land, is the binding constraint.** Every major province is short of grid capacity relative to requested data-centre load. Alberta's 1.2 GW interim connection allowance was fully allocated to two projects, and all remaining requests now proceed through a bring-your-own-generation framework (Sections 1 and 6).
- **The provinces have split into four distinct access models.** Alberta asks large loads to bring their own generation; Ontario screens projects for economic and sovereignty value under Bill 40; British Columbia runs a competitive allocation capped at 145 MW per project; and Quebec runs a selection process while seeking to nearly double the data-centre electricity rate. A national model that ignores these differences will misprice Canadian capacity (Section 8).
- **Credit quality, not price, now decides who gets capacity.** Hyperscalers are the anchor credit against which developers finance new campuses. Neoclouds increasingly face letters-of-credit requirements, prepayment and caps on their share of a campus, and in at least one reported case could not secure a lease despite offering a 15-year term and prepayment (Section 5).
- **Canada's AI-native capacity is being built on a three-partner model.** A Canadian landlord — often a telco — hosts a GPU operator, frequently a U.S. neocloud, which serves Canadian model companies, government and enterprise. Bell's 300 MW Sherwood campus near Regina, with CoreWeave and Cerebras as tenants, is the reference case (Sections 3 and 9).
- **Canada's two national telcos have chosen opposite models.** Both own their sites; the difference is who runs the GPUs. TELUS is an owner-operator, running its own sovereign GPU cloud inside its own data centres; Bell acts as landlord and sovereign AI channel, leaving GPU capital to its

tenants. The two models carry different risk profiles, and which proves more resilient if GPU prices fall or neocloud credit tightens is a central strategic question for the sector (Section 2.4).

- **AI workloads pull ordinary compute with them.** Inference and enterprise AI applications need their data and CPU-based systems nearby but not at GPU prices. This favours hyperscaler regions and mixed-density campuses that pair GPU halls with conventional colocation, and makes interconnection a competitive weapon for neoclouds (Section 2.6).
- **Liquid cooling has become the dividing line in facility design.** Average rack density rose from about 16 kW to 27 kW in a year, while Nvidia's next-generation racks draw up to 246 kW. Published estimates of liquid cooling's added capital cost run from roughly US\$2 million to more than US\$3 million per MW, and who pays for it is now a core lease negotiation (Section 7).
- **Sovereignty has become a commercial selection criterion.** Federal programs and Ontario's draft Playbook favour Canadian ownership or control; U.S. hyperscalers are responding with contractual data protections. Data residency, operational control and legal control are three different guarantees, and only the last is out of reach of foreign law (Section 9).
- **Regulators, not builders, set the next twelve months.** The Régie de l'énergie's ruling on Hydro-Québec's proposed data-centre rate, Ontario's final Data Centre Playbook, and the Alberta Electric System Operator's (AESO's) first bring-your-own-generation intake will shape where most new Canadian capacity is built through 2028 (Section 13).

How to use this Report

Readers assessing a specific transaction, landlord or tenant should pair Sections 1 and 3 (market sizing and the participants) with Section 5 (contracting and credit) and Section 12 (risk factors) as the core diligence starting point. Readers focused on siting and development — where power is available, on what terms, and how quickly — will find Sections 6 through 8 organized as a province-by-province reference. Section 9 sets out the federal and provincial sovereignty criteria now shaping funding and grid access, Section 13 identifies the catalysts most likely to reshape the market over the next 12 to 24 months, and Section 14 answers the questions most often raised by investors and boards. Section 15 and Appendix A provide a glossary and the full methodology and source list for readers who need to trace a specific figure back to its origin.

1. Market Overview and Size

Canada's data-centre market is growing at double-digit rates, but the more important fact for any investor or operator is the gap between what has been requested and what the provincial grids can serve. This section sizes the market, sets out the scale of the pipeline relative to grid capacity, and describes the metro and provincial geography in which that pipeline sits.

1.1 Market size and growth

Independent market research (Mordor Intelligence) values the Canadian data-centre market at approximately **US\$13.1 billion in 2026**, rising to a forecast **US\$25.1 billion by 2031** — a compound annual growth rate (CAGR) of about 13.9%. The hyperscale segment is the fastest-growing part of that market, forecast to rise from US\$3.8 billion to US\$10.2 billion over the same period (22.1% CAGR). Data-centre construction spending is forecast to grow from US\$3.7 billion to US\$6.3 billion. Exhibit 1 sets out these trajectories.

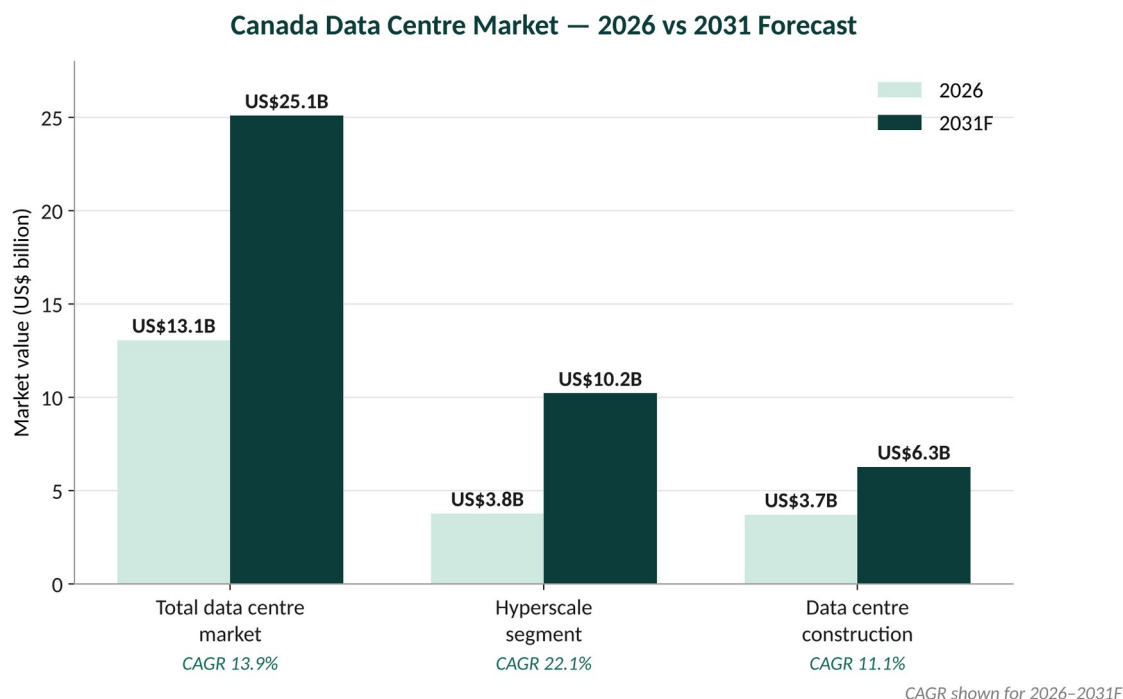


Exhibit 1. Canadian data-centre market, construction market and hyperscale segment, 2026 and 2031F, US\$ billion. Figures cover all data-centre types (enterprise, colocation, cloud and AI), not AI capacity alone. Source: Mordor Intelligence, as compiled by Brightlio (September 2026).

Two segmentation details matter for reading these figures. First, within the hyperscale segment, self-built facilities accounted for about **62%** of the market in 2025, while leased hyperscale colocation is forecast to grow faster, at roughly 23.5% a year through 2031. Second, campuses above 60 MW are the fastest-growing facility size class, while sites of 25–60 MW held about 41% of the hyperscale segment in 2025. Both details point in the same direction: the incremental market is shifting toward very large, leased campuses — the format in which most AI-native demand is likely to land (Section 4).

Reading the market data responsibly

In published market research, “hyperscale” describes a facility and customer size class, not only the hyperscalers themselves. A hyperscale colocation lease may be signed by a cloud provider, a neocloud, or an AI lab. Equally, the market and power figures in this section cover all data centres — enterprise, colocation, cloud and AI — because neither the research firms nor the provincial system operators separate AI capacity from the rest. The market-size figures in this section are third-party estimates rather than an audited industry census, and none of the hyperscalers or AI-native providers discussed in Section 3 publishes Canada-specific capacity or revenue. Readers underwriting a specific transaction should treat these figures as a starting reference point for diligence, not a substitute for it.

1.2 The pipeline is larger than the grids

Requested data-centre load is now a material multiple of available capacity in the two provinces with the most detailed public data. Exhibit 2 compares requested load with provincial peak demand.

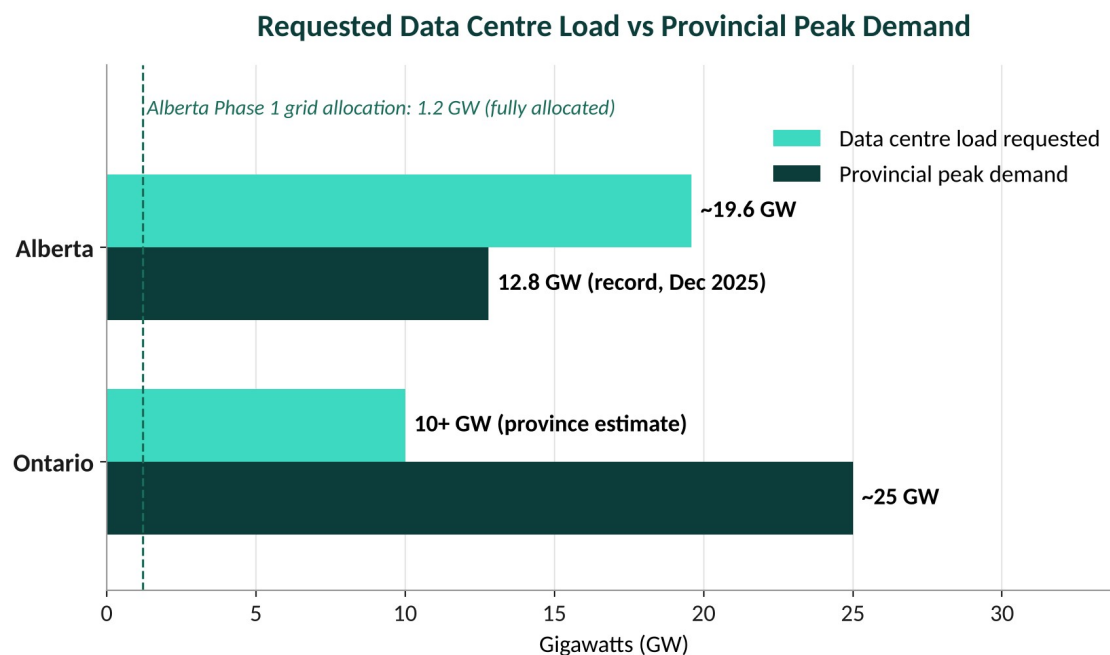


Exhibit 2. Requested data-centre load versus provincial peak demand, GW. Requested load covers all data-centre projects; the AI share is not separately reported. Sources: Government of Alberta (AESO connection list, July 30, 2026; record peak, December 11, 2025); Ontario Ministry of Energy and Mines via Dentons and Power Advisory (August–September 2026).

- **Alberta:** about 19,565 MW of new data-centre load was seeking connection on the AESO's list as of July 30, 2026, against a record provincial peak of 12,785 MW and installed generation of 23,393 MW. Industry briefings in September 2026 cited requests above 20,600 MW.
- **Ontario:** the province estimates connection requests could exceed 10,000 MW — roughly 40% of its recent peak demand of about 25,000 MW. The 2026 Annual Planning Outlook of the Independent Electricity System Operator (IESO), Ontario's grid operator, forecasts data-centre electricity consumption rising from 4.5 TWh in 2027 to 22.6 TWh by 2050 under its reference scenario.

- **Quebec:** data centres subject to Hydro-Québec's proposed new rate draw about 190 MW at peak, out of roughly 200 MW for the sector; the utility expects data-centre consumption to grow about sevenfold, to more than 1,000 MW, by 2035.

Other provinces do not publish a comparable list of all connection requests. What is public is the result of a capped call in British Columbia, a count of requests in Manitoba, and individual anchor projects elsewhere. These figures are not comparable with the Alberta and Ontario lists and are therefore not shown in Exhibit 2.

Province	Public data on data-centre load	Provincial peak	Largest project vs peak
British Columbia	15 applications totalling close to 800 MW in BC Hydro's 2026 Call for Demand, for up to 400 MW available; results pending. Data centres currently draw about 60 MW.	11,300 MW (January 2024)	Capped at 145 MW per project (~1%)
Saskatchewan	No public list of requests. Bell's AI campus near Regina is planned at about 300 MW at full build, supplied by SaskPower.	3,825 MW (2025–26)	Bell, ~300 MW (~8%)
Manitoba	13 connection requests (November 2025), no MW total disclosed. Convergence Compute (Jet.AI and Consensus Core) proposes a campus near Winnipeg of 100 MW initially, 500 MW at full build.	5,112 MW (January 2025)	Convergence Compute, 500 MW (~10%; proponent figure)
New Brunswick	Beacon Data Centers' proposed Lorneville campus: about 390 MW, roughly 190 MW from on-site gas and 200 MW from NB Power.	~3,030 MW (winter)	Beacon, ~200 MW from the grid (~7%)
Nova Scotia	No public figures; the premier would support a data centre only if the province could fully supply its energy.	—	—

Sources: BC Hydro (2026 Call for Demand; record peak, January 12, 2024); Business in Vancouver (March 2026); SaskPower Annual Report 2025–26; The Narwhal (November 2025; June 2026); DCD (December 2025); CBC News (Manitoba Hydro record, January 20, 2025); CTV News. Percentages are B² Advisory calculations against provincial peak.

The point for investors is one of scale. In the smaller provincial grids, a single AI campus is equivalent to 7–10% of provincial peak demand, so one project becomes a grid-planning and rate-setting event rather than an incremental load. This explains both Saskatchewan's direct utility arrangement with Bell and the public scrutiny of the Lorneville proposal in New Brunswick.

- **National:** a York University study counted 159 data-centre facilities proposed or under construction in Canada as of July 2026, 116 of them expected to exceed 100 MW, and a federal government document cited up to 20 GW of planned data-centre capacity.

The gap between requested and available capacity is itself the most important market signal in this Report. It means that the binding question for any Canadian project is no longer whether demand exists, but whether — and on what terms — a provincial system operator and regulator will allow it to connect. Sections 6 and 8 set out how each province is answering that question.

1.3 Metro and provincial geography

Canadian capacity has historically concentrated in two metros. **Toronto** hosts the largest enterprise and financial-services demand pool. **Montreal** hosts cloud regions for AWS, Google and Oracle and a large enterprise base, while the wider Quebec corridor has drawn energy-intensive workloads — historically cryptocurrency mining and, more recently, HPC and AI training — on the strength of low-cost hydroelectric power. To these, the current cycle has added three new geographies: the **Calgary-Edmonton corridor**, where Alberta's deregulated market and natural-gas supply support gigawatt-scale self-supplied campuses; **interior British Columbia**, where both national telcos are building hydro-backed sovereign AI capacity in Kamloops — TELUS in its existing data centre, Bell across three announced sites — with Bell also in nearby Merritt; and **Saskatchewan**, where Bell's Sherwood campus is the province's first large-scale AI facility.

Colocation has historically dominated the Canadian market by facility type — about 91% of market size in 2025 by Mordor's segmentation — and the market is moderately concentrated, with the top five operators holding just over 55% of installed capacity. That structure is now being challenged from two directions at once: hyperscalers self-building at gigawatt scale, and telcos entering as AI landlords.

1.4 Why Canada, and why now

- **U.S. constraint spillover:** as Northern Virginia, Santa Clara and other primary U.S. markets face power, land and permitting limits, developers are looking north for energized capacity.
- **Data-residency requirements:** federal government workloads, financial-services and public-sector buyers increasingly prefer or require data to stay in Canada, and Quebec's Law 25 requires a privacy impact assessment before personal information is transferred outside the province.
- **Sovereign AI policy:** the federal Canadian Sovereign AI Compute Strategy, a 2026 call for large-scale sovereign AI data centres, and provincial sovereignty criteria have turned domestic compute into an explicit policy objective (Section 9).
- **Clean and low-cost power:** Quebec and British Columbia hydro, Ontario's nuclear fleet, and Alberta's gas-plus-bring-your-own-generation model each offer a different but credible power proposition.
- **Climate:** cold winters reduce cooling energy for air-cooled facilities, although the advantage narrows as AI facilities move to liquid cooling (Section 7).



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